

EngageIQ — Smart Engagement Opportunity Scorer

Shivneel Prasad · BAX-423 Big Data · Spring 2026

Live URL: <https://engageiq-bax423-final.streamlit.app/>
GitHub: <https://github.com/shprasa/engageiq-bax423-final>

1. System architecture & pipeline

Pipeline: GitHub API + Hacker News API → in-app/Kafka stream queue → Bloom deduplication → Count-Min Sketch + HyperLogLog streaming metrics → DuckDB batch store → TF-IDF/SVD embeddings → cosine ANN retrieval → multi-stage ranker (relevance, health, visibility, effort, recency) → reinforcement learning policy (Thompson-sampling contextual bandit) from engage/skip/bookmark feedback → Streamlit dashboard with RL policy view, trend analytics, and CSV/PDF brief export. Offline snapshot: 10000 rows (1964 live API + 8036 synthetic backup), 15 domains. Live sources: {'hackernews': 1036, 'github': 928}.

2. BAX-423 techniques & benchmarks

Sketching: Bloom dedup detected 7 duplicate URLs; CMS/HLL power dashboard trend metrics. **Embeddings + ranking:** TF-IDF/SVD + NearestNeighbors; NDCG@10 reported per persona. **Reinforcement learning:** contextual multi-armed bandit (Thompson sampling) — state=user profile+history, actions=domain recommendations, rewards from engage/bookmark/skip. 60-round benchmark: NDCG@10 with RL 1.000 vs without 1.000; cumulative reward 60 vs 15; reward Δ(last 10)=+0.700; session gain=+0.000. Suggested actions: OpenAI when OPENAI_API_KEY set, else templates.

3. Persona results (top-10 checks)

Persona	NDCG@10	Checks	Overall
Sofia	0.933	GFI=7, C++/Rust=0	PASS
David	0.000	Infra=10/10	PASS
Lina	0.000	Visibility ≥ relevance	PASS
Raj	1.000	DevTools=10/10	PASS

4. Persona × capability pass/fail (required)

Persona	Multi-source ingest	Embeddings + ANN	Scoring + multi-stage	Adaptive learning / RL	Batch analytics + trend	Dashboard + brief expo
Sofia	PASS	PASS	PASS	PASS	PASS	PASS
David	PASS	PASS	PASS	PASS	PASS	PASS
Lina	PASS	PASS	PASS	PASS	PASS	PASS
Raj	PASS	PASS	PASS	PASS	PASS	PASS

5. Limitations

Synthetic backup rows (GitHub + HN only) ensure 10k offline grading. Streaming: in-app queue + optional Kafka (engageiq.opportunities topic). Embeddings use TF-IDF/SVD vs Sentence-BERT/FAISS for deploy reliability. Suggested actions are template-based (LLM optional). GH Archive/Kafka included as optional streaming extensions.